

BullLens: Smart Stock Tracker – Project Specification (Minor Project)

1. Introduction

BullLens is a local web-based application designed to help students and beginner investors track the stock market easily. The system provides near real-time stock prices, interactive historical charts, and simple machine learning insights through an easy-to-use interface. The project runs entirely on localhost and is designed for demonstration during academic evaluation such as a minor project viva. The goal is to combine web development, data visualization, API integration, and basic machine learning in a single educational platform.

2. Project Objectives

- Provide a simple platform to track stock prices.
- Visualize historical stock market data using interactive charts.
- Implement basic machine learning models to demonstrate trend analysis.
- Allow users to create and manage a personalized stock watchlist.
- Demonstrate integration of frontend, backend, database, and data analysis tools.
- Run entirely on a local machine for easy project demonstration.

3. System Overview

BullLens operates as a local web application. The frontend is built using HTML, CSS, and JavaScript. The backend uses Python with the Flask framework to handle requests, fetch stock data, and process machine learning tasks. Stock data is retrieved using the yfinance library, which provides historical and near real-time stock market information including Indian stocks such as RELIANCE.NS and TCS.NS. Processed data is visualized using Chart.js and stored locally in an SQLite database for watchlist management.

4. Key Features

- Near real-time stock price tracking.
- Historical stock data visualization.
- Interactive charts using Chart.js.
- Machine learning based trend analysis.
- Personalized watchlist feature.
- Beginner-friendly user interface.
- Fully local operation on localhost.

5. Technology Stack

Frontend:

- HTML5
- CSS3
- Bootstrap 5
- JavaScript
- Chart.js

Backend:

- Python 3
- Flask
- pandas
- scikit-learn

Data Source:

- yfinance library

Database:

- SQLite (local database file)

6. System Workflow

1. User opens the web application in the browser using localhost.
2. Frontend sends a request to the Flask backend.
3. Backend fetches stock data using the yfinance library.
4. Data is processed using pandas.
5. Machine learning models analyze historical patterns.
6. Watchlist information is stored in SQLite database.
7. Processed data is sent back to the frontend.
8. Charts and insights are displayed to the user.

7. Project Significance

BullLens provides an educational platform for students to understand stock market behavior using real data. The project demonstrates practical application of several computer science concepts including web development, data processing, machine learning, and database management. Because the system runs locally, it is simple to install, run, and demonstrate during academic presentations.

8. Future Scope

- Portfolio management features.
- Advanced technical indicators such as RSI and MACD.
- Deep learning based price prediction models.
- Mobile application version.

- Cloud deployment for public access.

9. Dataset Sources

Primary Data Source:

- yfinance Python library for live and historical stock data.

Optional Offline Training Data:

- Kaggle stock market datasets.
- Nifty 50 historical datasets.
- CSV exports generated from yfinance.